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CASE STUDY

FLOATING SOLAR

INDUSTRY	RENEWABLE ENERGY
CUSTOMER	TATA Power Renewable Limited
OBJECTIVE	SCADA & PPC of 70 MW Floating Solar
LOCATION	Kayakulam
PROJECT	TATA Kayakulam Floating Solar 70 MW

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CHALLENGES

- Designing solar SCADA system for floating solar applications (1st in India)
- Installing automation panels on floaters
- Calculating the Centre of Gravity (CG) to determine the sizing, weight and exact installation methods for the automation panels
- Protection against natural corrosion factors
- Skillset development for panel-to-panel interconnectivity

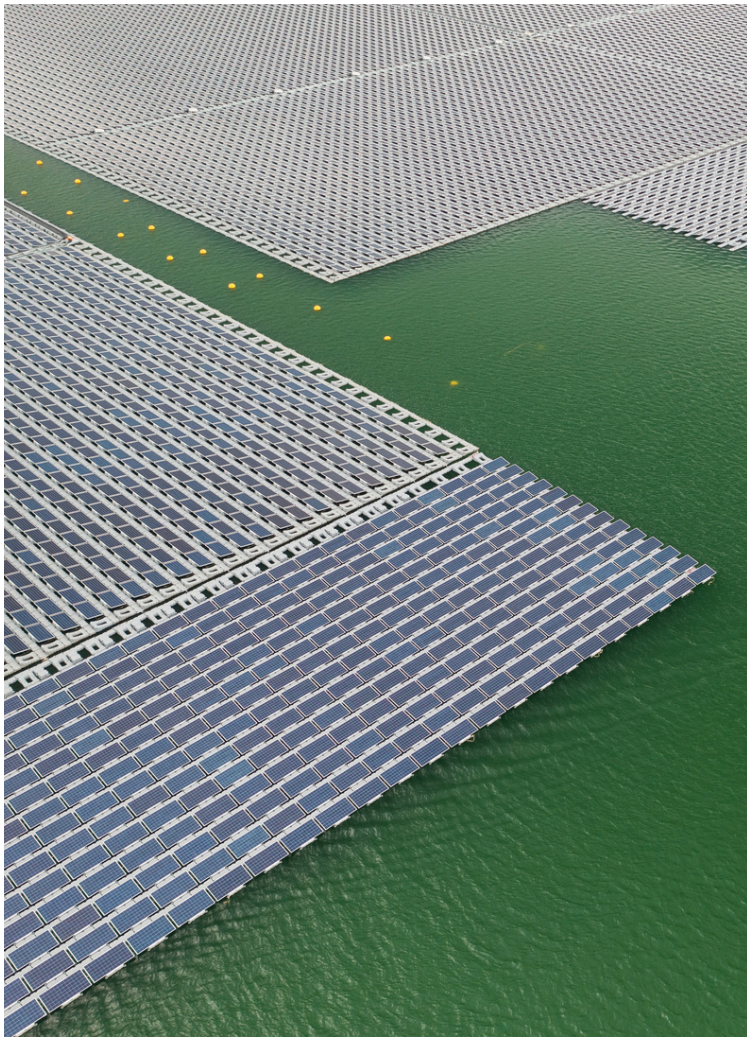




- Develop Master PPC to control additional BHEL 22 MW plant
- Redundant gateway for WRLDC data synchronization
- NTPC PI server integration over OPC-DA
- PPC Loop time optimization

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SOLUTION



✔ **Hot Standby PPC Controller:**

- Top Tier Hot Standby PPC Controller
- High processing power
- Highly reliable network configuration

✔ **Optimized Loop Time:**

- Optimized loop time enhancing the input scanning of PQM data
- PPC data processing
- Output writing to slave devices (inverters etc)

✔ **Customized PPC Logic:**

- Common station-based PQM
- Draw a reference from a single point (PQM)
- Distribute commands to the slave plant's PPCs (70 MW and 22 MW)

✔ **Electrical and Panel Design:**

- Threadbare study for centre of gravity calculation
- Floater max load capacity calculation
- Designed best in class electrical & panel solutions

✔ **Corrosion-Resistant Design:**

- Conformal coated controller and network devices
- Corrosion resistant paint applied on enclosures

✔ **Plant-to-Plant Communication:**

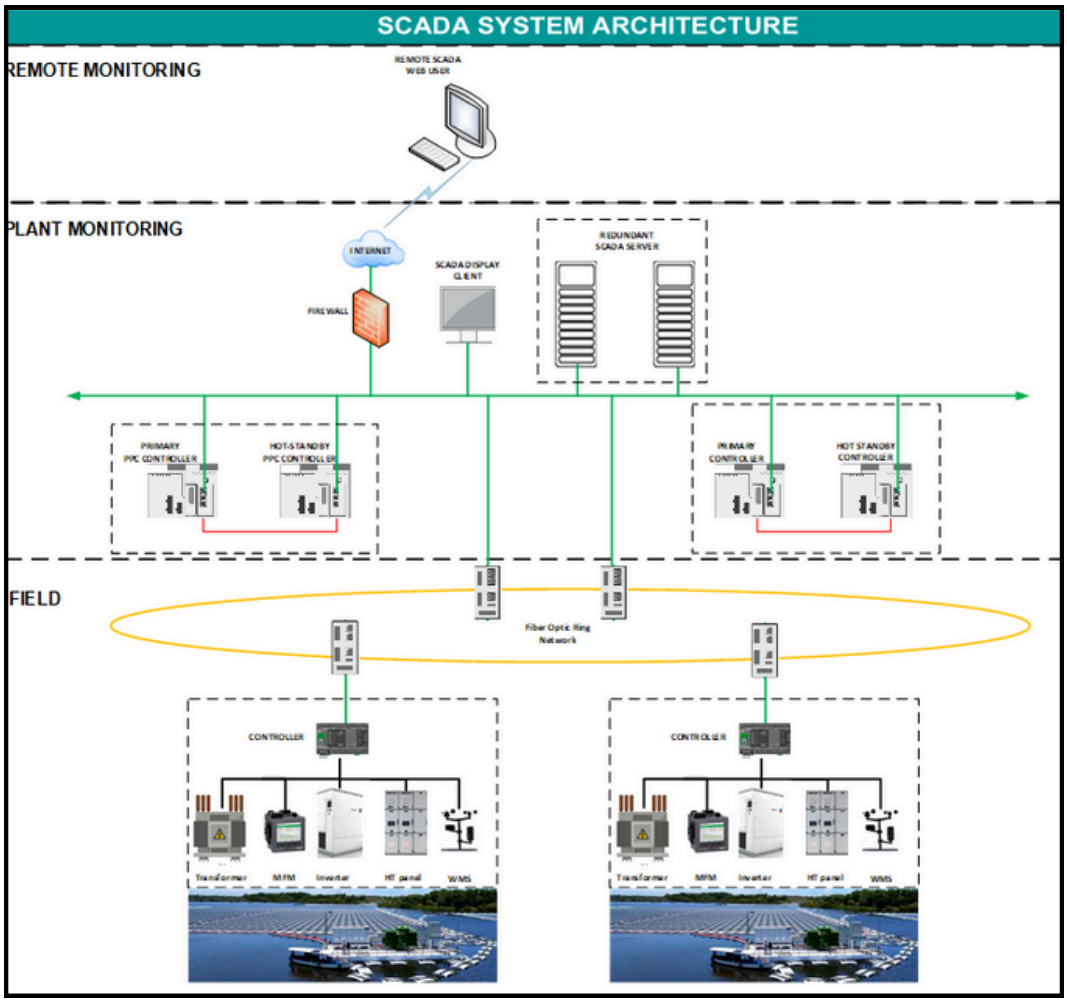
- Communication with BHEL SCADA (third-party)
- Inter-plant data sharing and control

✔ **NTPC PI Server Data Communication:**

- Communication with NTPC PI server over OPC-DA
- Through secure firewall policies



ARCHITECTURE





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VALUE CREATION

✔ **Comprehensive Compliance:**
- Ensure compliance with parameters viz. active power, reactive power, power factor, voltage, & frequency
- Single point of integration referencing at both plants



✔ **Master PPC to control multiple plants**

✔ **Fault-Tolerant Network:**
- Ensure high reliability and
- Availability of operations



✔ **Corrosion-Resistant System**

✔ **Reduced Downtime:**
- Redundant SCADA architecture
- Intuitive SCADA data screen
- Minimized downtime

21st Floor, "Yash Anant",
Opp. Old RBI, Ashram Road,
Ahmedabad - 380009,
Gujarat, India

+917969269000

info@adaptive-engg.com